POSITIVE ASYMPTOTIC BALL DENSITY OF DEAD ENDS IN AN INFINITE HYPERBOLIC CAYLEY GRAPH

ARGUS OPEN RESEARCH PROJECT

Abstract. Question 29 of Abért's problem list asks whether the density of dead ends in the Cayley graph of an infinite finitely generated group must tend to zero. Under the closed word-ball density convention, we document the counterexample of Calegari and Fujiwara (2015) on the hyperbolic $(2, 3, 7)$ triangle group.

1. Main Theorem

Let $\Gamma = \langle a, b \mid a^2 = b^3 = (ab)^7 = 1\rangle$ be the $(2, 3, 7)$ hyperbolic triangle group equipped with the symmetric generating set $S = \{a, b, b^{-1}\}$. An element $g \in \Gamma$ is a *dead end* with respect to S if $|gs|_S \leq |g|_S$ for all $s \in S$.

Theorem 1.1 (Calegari–Fujiwara)**.** *In the Cayley graph* $\mathrm{Cay}(\Gamma, S)$*, let* $B_S(n) = \{g \in \Gamma : |g|_S \leq n\}$ *denote the closed ball of radius n, and let* $D_S(n)$ *denote the subset of dead ends in* $B_S(n)$*. Then the asymptotic density exists and satisfies*

$$\lim_{n\to\infty} \frac{|D_S(n)|}{|B_S(n)|} \;\geq\; \frac{1}{64} \;>\; 0.$$

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